Zakhar Shumaylov

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Email: zakshum@gmail.com **Homepage**: zakobian.github.io **GitHub**: github.com/Zakobian
LinkedIn: linkedin.com/in/zakshumaylov **Scholar**: scholar.google.com/citations?user=EKV1lrAAAAAJ

Posts

Junior Research Fellow Oct 2026 – Oct 2030
Jesus College, University of Cambridge.

ProbAI EPSRC Doctoral Prize Fellow Oct 2026 – Oct 2027
University of Cambridge.

Education

University of Cambridge Cambridge, UK
PhD in Mathematics of Information 2022 – 2026
Thesis: "On the Limits of Machine Learning."
Supervised by: Prof Carola-Bibiane Schönlieb.
Awarded the *Trinity Henry Barlow Scholarship* (£81,000) at Christs College.
Funded by Christs College Warwick Postgraduate Studentship (£15,000) and CCIMI (£50,000).

University of Cambridge Cambridge, UK
Mathematics BA/MMath (1st Class/Distinction) 2018 – 2022
Awarded the *Cambridge Trust Scholarship* (£40,000) to read Mathematics at Churchill College.
Courses included: Quantum Field Theory, General Relativity, Statistical Field Theory, Black Holes, Cosmology.

Brighton College Brighton, UK
A-Level (5A*), STEP 2,3 (S,S) 2016 – 2018

Governor's Physics and Mathematics Lyceum 30 St-Petersburg, Russia
Year 9 - Year 11 (4.53/5) 2013 – 2016

Publications

[21] **AI models collapse when trained on recursively generated data**
Z. Shumaylov*, I. Shumailov*, Y. Zhao, Y. Gal, N. Papernot, R. Anderson
NATURE, 2024
Selected as cover and ranked 18th / 300k of articles.
Covered on the front page of New York Times.
One of the most influential articles of the year per State of AI report.
Publicity: New Scientist; Independent; The Atlantic; MIT tech; Financial Times; New York Times; Wall Street Journal; Bloomberg; The Register; AI Magazine; Cosmos

[20] **Muon is Not That Special: Random or Inverted Spectra Work Just as Well**
Z. Shumaylov, N. Da Costa, P. Zaika, B. Mucsányi, A. Massucco, Y. Gelberg, C. Schönlieb, Y. Gal, P. Hennig
UNDER REVIEW, 2026

[19] **Mathematics in the Age of Artificial Intelligence: A Primer on Key Topics in the Mathematical Sciences that Underpin AI**
D. S. Leslie, M. F. Anjos, O. Anosova, R. Bearon, M. Calder, J. A. Carrillo, A. Champneys, T. Coates, S. Cohen, J. Enright, A. Etheridge, D. Higham, S. Jaggi, V. Kurlin, R. Leese, V. Maguire Rajpaul, C. Marr, C. Nemeth, D. Nicolau, C. Offen, L. Pareschi, M. Pereyra, Y. Roudi, C. B. Schönlieb, **Z. Shumaylov**, D. Silvester, P. Tino, S. Veneziale, B. Walker, D. Woods, Y. Yu
Academy for the Mathematical Sciences, 2026

[18] **Adaptive Coordinate Transforms for Neural Operators**
C. Liu, Z. Li, G. Chen, **Z. Shumaylov**, Z. Deng, Q. Zhang, Z. Qiao, C. Schönlieb
Under review, 2026

[17] **When is a System Discoverable from Data? Discovery Requires Chaos**
Z. Shumaylov*, P. Zaika*, P. Scholl, G. Kutyniok, L. Horesh, C. Schönlieb
Under review, 2025

[16] **Diffeomorphism-Equivariant Neural Networks**
J. E. Oettinger, **Z. Shumaylov**, J. Bostelmann, J. Lellmann, C. Schönlieb
ICML, 2026

[15] **Generalized Lie Symmetries in Physics-Informed Neural Operators**
Z. Shumaylov*, A. X. Wang*, P. Zaika, F. Sherry, C. Schönlieb
Oral at SCML 2025; — **Oral & Best Paper Runner-up** at COLT TASC 2025

[14] **Learning Regularization Functionals for Inverse Problems: A Comparative Study**
J. Hertrich, H. S. Wong, A. Denker, S. Ducotterd, Z. Fang, M. Haltmeier, Z. Kereta, E. Kobler, O. Leong, M.S. Salehi, C. B. Schönlieb, J. Schwab, **Z. Shumaylov**, J. Sulam, G. S. Wache, M. Zach, Y. Zhang, M. J. Ehrhardt, S. Neumayer
Handbook of Numerical Analysis, Elsevier, 2026

[13] **Score-based pullback Riemannian geometry**
W. Diepeveen, G. Batzolis, **Z. Shumaylov**, C. Schönlieb
ICML, 2025

[12] **Lie Algebra Canonicalization: Equivariant Neural Operators under arbitrary Lie Groups**
Z. Shumaylov*, P. Zaika*, J. Rowbottom, F. Sherry, M. Weber, C. Schönlieb
ICLR, 2025

[11] **On Information Geometry and Iterative Optimization in Model Compression: Operator Factorization**
Z. Shumaylov, V. Tsiaras, Y. Stylianou
Under review, 2025

[10] **Symplectic Neural Flows for Modeling and Discovery**
P. Canizares, D. Murari, C. Schönlieb, F. Sherry, **Z. Shumaylov**
SIAM Journal on Scientific Computing, 2026

[9] **Hamiltonian Matching for Symplectic Neural Integrators**
P. Canizares, D. Murari, C. Schönlieb, F. Sherry, **Z. Shumaylov**
Oral at NeurReps, 2024

[8] **Benchmarking Learned Algorithms for Computed Tomography Image Reconstruction Tasks**
M. Kiss, A. Biguri, **Z. Shumaylov**, F. Sherry, J. Batenburg, C. Schönlieb, F. Lucka
Applied Mathematics for Modern Challenges, 2025

[7] **Data-driven approaches to inverse problems**
C. Schönlieb, **Z. Shumaylov**
CIME 2023 Notes, 2025

[6] **Weakly Convex Regularisers for Inverse Problems: Convergence of Critical Points & Primal-Dual Optimisation**
Z. Shumaylov, J. Budd, S. Mukherjee, C. Schönlieb
ICML, 2024

[5] **Data-Driven Convex Regularizers for Inverse Problems**
S. Mukherjee, S. Dittmer, **Z. Shumaylov**, S. Lunz, O. Öktem, C. Schönlieb
Oral at IEEE ICASSP, 2024

[4] **Provably Convergent Data-Driven Convex-Nonconvex Regularization**
Z. Shumaylov, J. Budd, S. Mukherjee, C. Schönlieb
Oral at NeurIPS Workshop on Deep Learning and Inverse Problems, 2023

[3] **Quantum Initial Conditions for Curved Inflating Universes**
Z. Shumaylov*, M. Letey*, F. Agocs, W. Handley, M. Hobson, A. Lasenby
Physical Review D, 2024

[2] **Primordial power spectra from k-inflation with curvature**
Z. Shumaylov, W. Handley
Physical Review D, 2022

[1] **Manipulating SGD with data ordering attacks**
I. Shumailov, **Z. Shumaylov**, D. Kazhdan, Y. Zhao, N. Papernot, M. Erdogdu, R. Anderson
NeurIPS, 2021

Experience

Google DeepMind — Mountain View, USA
Student Researcher — 08/25 – 01/26
Research on deep learning for inverse optical lithography.

Apple — Cambridge, UK
ML Research Intern — 12/24 – 08/25
ML research on model compression.

Apple Cambridge, UK
ML Research Intern 06/24 – 09/24
ML research on model compression using tensor networks.

University of Cambridge, DAMTP Cambridge, UK
Supervisor for Undergraduates and Postgraduates 10/22 – Present
Supervising undergraduate students in a variety of courses.
(2022/2023): Part IA Vectors and Matrices: 18 students (48h);
(2023/2024): Summer Project Supervision: 2 students;
(2024/2025): MPhil and Summer Project Supervision: 2 + 2 students.

GSK Cambridge, UK
Project Collaboration 06/22 – 09/22
Project collaboration on "Self-discovery of mechanistic equations for a data-driven smart simulator" as part of CMI programme with Dr Matthieu Duvinage.

Ryff AI Cambridge, UK
Summer Research Intern 07/22 – 09/22
Work under supervision of Dr Mike Roberts. During the internship I worked on the problem of unsupervised video motion segmentation. During the project, I used variational and learned methods from the optical flow literature for foreground-background separation using motion signals.

University of Cambridge, Institute of Astronomy Cambridge, UK
Summer Internship Programme 08/21 – 09/21
Work under supervision of Dr Amy Bonsor (IoA): "Gas disk imaging around white dwarves". During the internship I investigated gas disk light curve imaging around white dwarves, by modelling gas geometry. Funded by the Institute of Astronomy.

University of Cambridge, KICC Cambridge, UK
Summer Research Intern 06/21 – 08/21
Work under supervision of Dr Will Handley (KICC): "Primordial power spectra from k-inflation with curvature". During the internship I investigated the problem of interplay between inflationary sound speed and primordial curvature using analytical approximations. Funded by the CMP.

University of Cambridge, DAMTP Cambridge, UK
Summer Research Assistant 06/20 – 09/20
Work under supervision of Prof Carola Schonlieb (DAMTP), Prof Ozan Oktem (KTH) and Prof Par Kurlberg (KTH): "3DEM: Representation of atomic models". During the internship I investigated the problem of protein fitting inside of atomic volumes acquired via cryo electron microscopy. During the project I used learned techniques and variational methods to obtain protein reconstructions. Funded by the CSRIM.

University of Cambridge, DAMTP Cambridge, UK
Summer Research Assistant 06/19 – 09/19
Work under supervision of Prof Carola Schonlieb (DAMTP). During the internship I worked primarily in the field of inverse problems. In particular, I researched how Deep Learning can be used to help solve physics-based inverse imaging problems. This led to a joint work "Learned convex regularizers for inverse problems". Funded by the CSRIM and the Tizard Fund.

Cambridge Coding Academy Cambridge, UK
Teaching Assistant 07/18
Supporting and leading coding sessions of the 'Coding++' course, covering the basics of AI using python and the pygame library.

Brighton College Brighton, UK
After-school Teaching Assistant 09/17 – 06/18
Tutoring Year 9 - Year 13 students during after-school Mathematics classes.

University of Sussex Brighton, UK
Research Assistant to Professor Madzvamuse 07/17 – 08/17
I reviewed and extended the one-dimensional cell model of Shenoy(2016) by modelling cell contractility and strain with partial differential equations in Matlab.

Subject Olympiads

British Physics Olympiad Round 2 (Gold Award, Top 15) 2018
Invited to University of Oxford Training Camp to compete for UK IPhO team.

British Astronomy and Astrophysics Olympiad (Gold Award) 2018

British Physics Olympiad Round I (Gold Award, Top 50) 2017

British Mathematics Olympiad Round I (Certificate of Distinction) 2017

British Physics Olympiad Round I & AS Physics Challenge (Gold Awards) 2016

Senior Mathematics Challenge (Gold Award, 100%) 2016

School Mathematics Olympiad (Winner of inter-school team challenge) 2016

Russian Computer Science & Physics Olympiads (Winner of district challenges) 2015

Russian Computer Science Olympiad (Winner of district challenge) 2014

Awards & Fellowships

EAIP PhD Thesis Award (€1,000) 2026
Eurasian Association for Inverse Problems, awarded for outstanding PhD thesis.

G-Research PhD Prize, 2nd place (£6,000) 2026
Awarded by G-Research for outstanding PhD research.

Trinity Henry Barlow Scholarship (£81,000) 2022
Awarded to pursue PhD in Mathematics of Information at University of Cambridge.

Cambridge Christs Bursary (£15,000) 2022
Awarded to pursue PhD in Mathematics of Information at University of Cambridge.

CCIMI (£50,000) 2022
Awarded to pursue PhD in Mathematics of Information at University of Cambridge.

C.I.M.E. full grant (€ 1,000) 2023
Awarded to attend CIME school on 'Machine Learning: From Data to Mathematical Understanding'.

Churchill College Prize Scholarship (£120) 2021
Awarded in recognition of excellent academic performance.

Churchill College Honorary Scholarship (£100) 2020
Awarded in recognition of excellent academic performance.

Churchill College Prize Scholarship (£120) 2019
Awarded in recognition of excellent academic performance.

Cambridge Trust Scholarship (£40,000) 2018
Awarded to read Mathematics at University of Cambridge.

Brighton College Governors Award for Independent Study (£500) 2018
For a piece of work outside of the A-Level curriculum.

Brighton College Physics Prize: Bayliss-Smith prize 2018
Prize to recognise sustained excellence and scientific endeavor.

Brighton College Science Essay Competition 2018
Winning essay: "The Tale of Cell Modelling".

Brighton College Science Prize: Newton's Cup 2017
Prize to recognise sustained excellence and scientific endeavor.

Brighton College Science Essay Competition 2017
Winning essay: "Brief History of Exoplanets".

Invited Talks

SIAM Conference on Optimization 2026 (Edinburgh, UK):
Invited to present on "Learning Regularization Functionals for Inverse Problems," "Structure Preservation in Inverse Problems," and "When is a System Discoverable from Data?"

ProbAI Retreat (Edinburgh, UK):
Invited to present on "When is a System Discoverable from Data? Discovery Requires Chaos."

REMODEL Conference, NTNU (Trondheim, Norway):
Invited to present on "When is a System Discoverable from Data? Discovery Requires Chaos."

NeurIPS @ Cambridge (Cambridge, UK):
Invited to present on "When is a System Discoverable from Data?"

Machine Learning Journal Club (MLJC) at Gatsby UCL (London, UK):
Invited to present on "The Future of Synthetic Data: Model Collapse and Equivariant Neural Operators."

Workshop on Lie Groups and Symmetry at The Alan Turing Institute (London, UK):
Invited to present on "Symmetries in Neural O/PDE Solvers."

Maths4DL Conference on Inverse Problems and Deep Learning (Bath, UK):
Invited to present on "Symmetries in Neural PDE Solvers."

BAMC 2025 (Exeter, UK):
Invited to present on "Convergent Data-Driven Regularisation in Inverse Problems."

NUS (Singapore):
Invited to present on "Symmetries in Neural O/PDE Solvers."

Harvard University (Boston, USA):
Invited to present on "Symmetries in Neural O/PDE Solvers."

Christs College (Cambridge, UK):
Invited to present on "AI Models collapse when trained on recursively generated data" as part of ERSS series.

TU Berlin (Berlin, Germany):
Invited to present on "AI Models collapse when trained on recursively generated data."

Tübingen AI Center (Tübingen, Germany):
Invited to present on "The Future of Synthetic Data: Model Collapse and Equivariant Neural Operators."

Oberwolfach workshop on "Deep Learning for PDE-based Inverse Problems" (Oberwolfach, Germany):
Invited to present on "Lie Algebra Canonicalization: Equivariant Neural Operators under arbitrary Lie Groups."

European Congress of Mathematics 2024 (Seville, Spain):
Invited to present on "Weakly convex regularizers in inverse problems."

KTH SciML workshop (Stockholm, Sweden):
Invited to present on "Weakly convex regularizers in inverse problems."

AI Precision Health Institute (Hawaii, USA):
Invited to present on "What happens if we use synthetic data without any curation."

SIAM Imaging 2024 (Atlanta, USA):
Invited to present on "Weakly convex regularizers in inverse problems."

IEEE ICASSP 2024 (Seoul, South Korea):
Invited to present on "Data-Driven Convex Regularizers for Inverse Problems."

NeurIPS @ Cambridge (Cambridge, UK):
Presented on "The Curse Of Recursion: Generated Data Makes Models Forget."

Workshop: Integrating acquisition and AI in tomography (Leiden, Netherlands):
Presented on "Learned reconstruction methods in inverse problems."

ICIAM 2023 (Tokyo, Japan):
Invited to present on "Learned weakly convex regularizers in inverse problems."

C.I.M.E. School on 'Machine Learning: From Data to Mathematical Understanding' (Cetraro, Italy):
Received full grant and prepared lecture notes to be published in the C.I.M.E. Springer series.

Service

Reviewing for Conferences: ICML, ICLR, NeurIPS, IEEE ICASSP, AAAI

Reviewing for Workshops: SLLM, NeurReps, GRaM, WSS

Reviewing for Journals: IMA Journal of Numerical Analysis, Philosophical Transactions of the Royal Society A, IEEE Transactions on Computational Imaging, IEEE Transactions on Information Theory, Foundations of Data Science, ACM Transactions on Intelligent Systems and Technology, Journal of Computational Physics, Numerical Functional Analysis and Optimization

Organization: NeurIPS @ Cambridge meetups (2023, 2024). Member of IPYA (Inverse Problems Young Academics).

Leadership Positions: Treasurer and Membership officer at Cambridge University Astronomical Society; Deputy Head of School House at Brighton College; Founder and President of Brighton College STEM Society; Leader of the House Chess Team at Brighton College.

Technical Skills

Programming Languages: Python, C
Software Packages: pyTorch, jax, odl, Matlab
Other Tools: Maple, Mathematica, LaTeX
OS & Computing: Linux, MacOS, unix, bash, slurm, HPC, vim